Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Applications of Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Derivatives · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Step 1 · Initial Verification: As $x \to 0$, numerator $0 - 0 = 0$ and denominator $0^3 = 0$. Form $\frac{0}{0}$. Differentiating numerator and denominator: $\lim_{x \to 0} \frac{1 - \frac{1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{\frac{(1+x^2)-1}{1+x^2}}{3x^2} = \lim_{x \to 0} \frac{x^2}{3x^2(1+x^2)}$. Step 2 · Algebraic Simplification: For $x \neq 0$, cancel the common factor x^2: $= \lim_{x \to 0} \frac{1}{3(1+x^2)} = \frac{1}{3(1+0)} = \mathbf{\frac{1}{3}}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

Compute gradient: $ablaf = \langle 6xy, 3x^2 - 3y^2 \rangle$. At $(1, 2)$: $ablaf(1, 2) = \langle 6(1)(2), 3(1)^2 - 3(2)^2 \rangle = \langle 12, 3 - 12 \rangle = \langle 12, -9 \rangle$. Maximum rate of change is the gradient magnitude: $|ablaf(1, 2)| = \sqrt{12^2 + (-9)^2} = \sqrt{144 + 81} = \sqrt{225} = 15$ The direction of maximum increase is the normalized gradient: $\mathbf{u} = \frac{ablaf}{|ablaf|} = \frac{1}{15}\langle 12, -9 \rangle = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

Complete the square: $y = (x - 3)^2 - 9 + 5 = (x - 3)^2 - 4$. The vertex of the parabola is $(3, -4)$. To map $y = (x - 3)^2 - 4$ back to $y = x^2$, we must translate the graph 3 units to the left and 4 units upwards. In vector notation, this is a single translation by $(-3; 4)$, or two sequential transformations: (1) Translation by vector $(-3; 0)$ (3 units left), and (2) Translation by vector $(0; 4)$ (4 units up).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

Take natural logarithms of both sides: $\ln y = \ln(x^{\sin x}) = \sin x \cdot \ln x$. Differentiate implicitly with respect to x using the Product Rule on the right: $\frac{1}{y}\frac{dy}{dx} = (\cos x)(\ln x) + (\sin x)\left(\frac{1}{x}\right)$. Multiply both sides by $y = x^{\sin x}$: $\frac{dy}{dx} = x^{\sin x}\left(\cos x \ln x + \frac{\sin x}{x}\right)$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

Express $y = (x^4 + 3x^2 + 7)^{1/2}$. By the Chain Rule: $y' = \frac{1}{2}(x^4 + 3x^2 + 7)^{-1/2} \cdot \frac{d}{dx}(x^4 + 3x^2 + 7) = \frac{4x^3 + 6x}{2\sqrt{x^4 + 3x^2 + 7}} = \frac{2x^3 + 3x}{\sqrt{x^4 + 3x^2 + 7}}$. Evaluate at $x = 1$: $y'(1) = \frac{2(1)^3 + 3(1)}{\sqrt{1^4 + 3(1)^2 + 7}} = \frac{5}{\sqrt{11}} = \mathbf{\frac{5\sqrt{11}}{11}}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.

(a) $y = f(x + 3) + 2$ translates 3 units left and 2 units up: $(4 - 3, -6 + 2) = (1, -4)$. (b) $y = 3f(2x)$ stretches horizontally by $\frac{1}{2}$ and vertically by 3: $\left(4 \times \frac{1}{2}, -6 \times 3\right) = (2, -18)$. (c) $y = -f(-x)$ reflects in both axes (point reflection through the origin): $(-4, -(-6)) = (-4, 6)$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 1 · Foundations Find $f_x(2,1)$ and $f_y(2,1)$ for the function $f(x,y)=\ln(x^2+3y^2)$.

Compute partial derivatives via Chain Rule: $f_x=\frac{2x}{x^2+3y^2}$, $f_y=\frac{6y}{x^2+3y^2}$ Substitute $x=2$ and $y=1$: $x^2+3y^2=2^2+3(1^2)=4+3=7$ $f_x(2,1)=\frac{2(2)}{7}=\frac{4}{7}$, $f_y(2,1)=\frac{6(1)}{7}=\frac{6}{7}$

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Indices, Surds & Quadratics · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Differentiation & Integration · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

[Simultaneous Equations & Inequalities · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.01 Further Integration Techniques**

[Trigonometry · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 5.02 Differential Equations**

[Applications of Integration · Practice 4] If the governing relation for this system is $g(t)=6t+6$, evaluate $g(10)$ and find the inverse input t when $g(t)=72$.

Substitute $t=10$: $g(10)=6(10)+6=\mathbf{66}$. Setting $6t+6=72$ gives $6t=66\Rightarrow t=\mathbf{11}$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Matrices (AI HL)

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.03 Markov Chains (AI HL)

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the domain and range of the principal arcsine function $y = \arcsin x$?

a. **Domain:** $[-1, 1]$**; Range:** $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ ✓

b. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

c. Domain: $[-1, 1]$; Range: $[0, \pi]$

d. Domain: $(-\infty, \infty)$; Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

To make sine invertible, its domain is restricted to $[-\pi/2, \pi/2]$ with range $[-1, 1]$. Thus $\arcsin x$ has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the domain and range of $y = \arccos x$?

a. **Domain:** $[-1, 1]$**; Range:** $[0, \pi]$ ✓

b. Domain: $[-1, 1]$; Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

d. Domain: $[0, 1]$; Range: $[0, \pi/2]$

The cosine function is restricted to $[0, \pi]$ where it is strictly decreasing. Therefore $\arccos x$ maps $[-1, 1]$ onto $[0, \pi]$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the horizontal asymptotes of the graph of $y = \arctan x$?

a. $y = \frac{\pi}{2}$ **and** $y = -\frac{\pi}{2}$ ✓

b. $y = \pi$ and $y = 0$

c. $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

d. $y = 1$ and $y = -1$

As $x \to \infty$, $\arctan x \to \frac{\pi}{2}$; as $x \to -\infty$, $\arctan x \to -\frac{\pi}{2}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

a. $\frac{3}{\sqrt{1 - 9x^2}}$ ✓

b. $\frac{1}{\sqrt{1 - 9x^2}}$

c. $\frac{3}{\sqrt{1 - 3x^2}}$

d. $\frac{3}{1 + 9x^2}$

By the chain rule: $\frac{d}{dx}[\arcsin(u)] = \frac{u'}{\sqrt{1 - u^2}} = \frac{3}{\sqrt{1 - (3x)^2}} = \frac{3}{\sqrt{1 - 9x^2}}$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

a. $\frac{2}{4+x^2}$ ✓

b. $\frac{1}{1+x^2/4}$

c. $\frac{1}{4+x^2}$

d. $\frac{2}{\sqrt{4-x^2}}$

$\frac{d}{dx}[\arctan(x/2)] = \frac{1/2}{1+(x/2)^2} = \frac{1/2}{1+x^2/4} = \frac{1/2}{(4+x^2)/4} = \frac{2}{4+x^2}$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

a. $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$ ✓

b. $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

c. $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

d. They are reciprocals of each other

Because $\arcsin x + \arccos x = \frac{\pi}{2}$ for all $x \in [-1, 1]$, differentiating both sides yields $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x]$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

a. 1 ✓

b. 0

c. $\pi/2$

d. Undefined

$y'(x) = \frac{1}{\sqrt{1-x^2}}$. At $x = 0$, $y'(0) = \frac{1}{\sqrt{1-0}} = 1$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?

a. **The slope approaches $+\infty$ (vertical tangent) ✓**

b. The slope approaches 0

c. The slope approaches 1

d. The slope approaches -1

As $x \to 1^-$, $\sqrt{1-x^2} \to 0^+$, so $y'(x) = \frac{1}{\sqrt{1-x^2}} \to +\infty$, resulting in a vertical tangent at the boundary.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

a. $y - \frac{\pi}{4} = \frac{1}{2}(x-1)$ ✓

b. $y - \frac{\pi}{4} = 1(x-1)$

c. $y = \frac{1}{2}x + \frac{\pi}{4}$

d. $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$

Point is $(1, \arctan 1) = (1, \pi/4)$. Derivative is $y'(1) = \frac{1}{1+1^2} = \frac{1}{2}$. Tangent line: $y - \frac{\pi}{4} = \frac{1}{2}(x-1)$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $g(x) = f(x+1) - 5$ is obtained from $f(x)$ by:

a. Shifting 1 units left and 5 units down

b. **Shifting 1 units right and 5 units down ✓**

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Replacing x with $x - 1$ shifts horizontally by 1 (right if positive); adding -5 shifts vertically by -5.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k is $f(x) = -2x + k$ (for $x < 5$) and $f(x) = x^2$ (for $x \geq 5$) continuous at $x = 5$?

a. 36

b. 34

c. -35

d. **35** ✓

Continuity requires $-2x + k$ at $x = 5$ to equal $(5)^2$: $-10 + k = 25$, so $k = 35$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If $f(x) = \sqrt{x - 5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \leq x \leq 3$

b. $x \leq -3$ **or** $x \geq 3$ ✓

c. $x \geq 3$

d. All real numbers

Need $x^2 \geq 9$, so $|x| \geq 3$, giving $x \leq -3$ or $x \geq 3$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. **-5** ✓

Since $x = 3 \geq 3$, use $f(x) = -2x + 1$: $f(3) = -2(3) + 1 = -5$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. **Odd** ✓

c. Neither

d. Cannot be determined

Every term has odd degree, so $f(-x) = -f(x)$: odd.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ **or** $5 < k < 11$ ✓

b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

1. **Find $f^{-1}(x)$ and its domain:** Completing the square gives $f(x) = (x - 3)^2 + 4$. For the domain $x \geq 3$, $f(x)$ is strictly increasing with range $[4, \infty)$. Setting $y = (x - 3)^2 + 4$ with $x \geq 3$ yields $x - 3 = \sqrt{y - 4}$, so $f^{-1}(x) = 3 + \sqrt{x - 4}, \quad \text{with domain } x \geq 4$. 2. **Set up the equation $f^{-1}(x) = |2x - k|$:** Let $u = \sqrt{x - 4} \geq 0$, which implies $x = u^2 + 4 \geq 4$. The equation becomes $3 + u = |2(u^2 + 4) - k| = |2u^2 + 8 - k|$. 3. **Analyz.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The function f is defined for $x \in \mathbb{R}$ by $f(x) = 3x + k,$ where k is a constant. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - 4|$. Given that $f^{-1}(4) = 2$, find the value of k and all real solutions to the equation $f^{-1}(x) = g(x)$.

a. $k = -2;\ x = \frac{5}{2},\ x = 7$ ✓

b. $k = -2; \; x = 7$)

c. $k = 2;\ x = \frac{7}{2},\ x = 5$

d. $k = -2;\ x = -\frac{5}{2},\ x = 7$

1. **Find the parameter k:** By the definition of an inverse function, $f^{-1}(4) = 2 \Leftrightarrow f(2) = 4$. Substituting $x = 2$ into $f(x)$: $f(2) = 3(2) + k = 6 + k\ 6 + k = 4 \Rightarrow k = -2$ 2. **Determine the inverse function $f^{-1}(x)$:** With $k = -2$, $f(x) = 3x - 2$. Setting $y = 3x - 2$ and solving for x: $3x = y + 2 \Rightarrow x = \frac{y + 2}{3}$ Thus, $f^{-1}(x) = \frac{x + 2}{3}$. 3. **Solve the equation $f^{-1}(x) = g(x)$:** $\frac{x + 2}{3} = |x - 4| \Leftrightarrow x + 2 = 3|x - 4|$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What are the values of x that satisfy the equation $|3x-2|=2|x+1|$?

a. $x=0$ **and** $x=4$ ✓

b. $x=0$ and $x=-4$

c. $x=\frac{1}{5}$ and $x=4$

d. $x=\frac{1}{4}$ and $x=\frac{3}{2}$

To solve the modulus equation $|3x-2|=2|x+1|$, we can either consider the linear cases or square both sides. **Method 1: Considering linear cases** The equation $|A|=|B|$ (where $B=2(x+1)$) is equivalent to $A=B$ or $A=-B$: - Case 1: $3x-2=2(x+1)\Rightarrow 3x-2=2x+2\Rightarrow x=4$ - Case 2: $3x-2=-2(x+1)\Rightarrow 3x-2=-2x-2\Rightarrow 5x=0\Rightarrow x=0$ **Method 2: Squaring both sides** Since both sides are non-negative, we can square both sides.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Matrices (AI HL)

The function f is defined for $x\in\mathbb{R}, x\neq 3$ by $f(x)=\frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x\neq 2$?

a. $\frac{3x+1}{x-2}$ ✓

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

To find the inverse function $f^{-1}(x)$, set $y=f(x)$ and rearrange to express x in terms of y: $y=\frac{2x+1}{x-3}$ Multiply both sides by $x-3$: $y(x-3)=2x+1$ $yx-3y=2x+1$ Collect all terms containing x on one side and constant/independent terms on the other: $yx-2x=3y+1$ Factor out x: $x(y-2)=3y+1$ Divide by $y-2$: $x=\frac{3y+1}{y-2}$ Interchanging the variables gives: $f^{-1}(x)=\frac{3x+1}{x-2}$ Common Misconceptions: - Choice.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. **Graph B ✓**

c. Graph C

d. Graph D

The graph of the inverse function $y = f^{-1}(x)$ is obtained by reflecting the graph of $y = f(x)$ across the line $y = x$. The provided graph of $f(x)$ is a parabola opening upwards with its vertex at $(2, -1)$. Graph A shows a function that is not a reflection of $f(x)$ across $y = x$. Graph B correctly shows the reflection of $y = f(x)$ across the line $y = x$. For example, the point $(0, 3)$ on $f(x)$ corresponds to the point $(3, 0)$ on $f^{-1}(x)$. The vertex $(2, -1)$ on $f(x)$ corresponds to the point.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.03 Markov Chains (AI HL)

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1 - x^2}$ ✓

b. $\frac{1}{\sqrt{1 - x^2}}$

c. $1 - x^2$

d. $\sqrt{x^2 - 1}$

Let $\theta = \sin^{-1}(x)$, so $\sin\theta = x$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, where $\cos\theta \geq 0$. Then $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - x^2}$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Proof by Mathematical Induction

A ladder 10 m long leans against a wall; the base slides away at 1 m/s. How fast is the top sliding down when the base is 6 m from the wall?

a. $\frac{3}{4}$

b. $-\frac{3}{8}$

c. $-\frac{3}{4}$ ✓

d. $-\frac{3}{2}$

$x^2 + y^2 = 10^2$. At $x = 6$: $y = \sqrt{10^2 - 6^2} = 8$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{3}{4}$ m/s.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

If $\sin\theta = -\frac{20}{29}$ and θ is in Quadrant 3, find $\cos\theta$.

a. $\frac{29}{21}$

b. $\frac{21}{29}$

c. $-\frac{21}{29}$ ✓

d. $\frac{20}{29}$

By the Pythagorean identity, $\cos\theta = \pm\sqrt{1-\sin^2\theta} = \pm\frac{21}{29}$. In Quadrant 3, cosine is negative, so $\cos\theta = -\frac{21}{29}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. **Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$** ✓

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (restricted principal output).

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.04 Partial Fractions

Implicit function theorem for optimization: constrained critical points satisfy $abla_{\mathbf{x}} L = 0$ where $L = f - \lambda g$. Name of λ:

a. **Lagrange multiplier** ✓

b. eigenvalue

c. gradient

d. curvature

Applying Application of Derivatives principles, λ = Lagrange multiplier. Therefore, the verified correct option is Lagrange multiplier.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

If $f'(x) = (x - 3)$, on which interval is f increasing?

a. $(3, \infty)$ ✓

b. \{3\}

c. $(-\infty, \infty)$

d. $(-\infty, 3)$

Applying Application of Derivatives principles, $f'(x) > 0$ when $x - 3 > 0$, i.e. $x > 3$. Therefore, the verified correct option is $(3, \infty)$.

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